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Homework 3

4.5

Suppose that zero interest rates with continuous compounding are as follows:

Maturity (months)	Rate (% per annum)
3	8.0
6	8.2
9	8.4
12	8.5
15	8.6
18	8.7

Calculate forward interest rates for the second, third, fourth, fifth, and sixth quarters.

We know that the forward rate for the period between T_1 and T_2 implied by today's curve is $R_F = \frac{R_2T_2 - R_1T_1}{T_2 - T_1}$, so we calculate R_F accordingly, with each quarter being $0.25T$.

Quarter	Forward interest rate
2	$\frac{0.082(0.5)-0.08(0.25)}{0.25} = \frac{0.041-0.02}{0.25} = \frac{0.021}{0.25} = 0.084$, or **8.4%**
3	$\frac{0.084(0.75)-0.082(0.5)}{0.25} = \frac{0.063-0.041}{0.25} = \frac{0.022}{0.25} = 0.088$, or **8.8%**
4	$\frac{0.085(1.00)-0.084(0.75)}{0.25} = \frac{0.085-0.063}{0.25} = \frac{0.022}{0.25} = 0.088$, or **8.8%**
5	$\frac{0.086(1.25)-0.085(1.00)}{0.25} = \frac{0.1075-0.085}{0.25} = \frac{0.0225}{0.25} = 0.09$, or **9.0%**
6	$\frac{0.087(1.50)-0.086(1.25)}{0.25} = \frac{0.1305-0.1075}{0.25} = \frac{0.023}{0.25} = 0.092$, or **9.2%**

4.8

What does duration tell you about the sensitivity of a bond portfolio to interest rates? What are the limitations of the duration measure?

Duration measures the **bond portfolio's value sensitivity relative to to interest rate changes on the yield curve**. By definition, we use the duration to calculate the amount that the value of the portfolio changes: the percentage decrease in the value of the bond portfolio equals the duration multiplied by the size of the shift on the yield curve. The duration is limited as a measure in that the shift on the yield curve must **both small and parallel**, with all maturities moving by the same amount.

4.9

What rate of interest with continuous compounding is equivalent to 15% per annum with monthly compounding?

We solve using the corresponding equation for continuous compounding, with $R_m = 0.15, m = 12$:

$$e^{R_c} = (1 + \frac{R_m}{m})^m$$
$$e^{R_c} = (1 + \frac{0.15}{12})^{12}$$
$$e^{R_c} = 1.0125^{12}$$
$$R_c = \ln(1.0125^{12})$$
$$R_c = 12 \ln(1.0125)$$
$$R_c \approx 0.14907023988$$

which indicates that the rate is about $\boxed{\mathbf{14.91\%}}$.

4.11

Suppose that 6-month, 12-month, 18-month, 24-month, and 30-month zero rates are 4%, 4.2%, 4.4%, 4.6%, and 4.8% per annum with continuous compounding respectively. Estimate the cash price of a bond with a face value of 100 that will mature in 30 months and pays a coupon of 4% per annum semiannually.

We need to discount each cash flow at the matching zero rate. We know that the coupon pays $\frac{0.04 \cdot 100}{2} = = \$2$ every 6 months, and then $100 + 2 = \$102$ at the last payment segment. We therefore would calculate the price like so:

$$P = 2e^{-0.04(0.5)} + 2e^{-0.042(1.0)} + 2e^{-0.044(1.5)} + 2e^{-0.046(2)} + 102e^{-0.048(2.5)}$$

$$= 1.96039734661 + 1.91773956114 + 1.87226172858 + 1.82421029909 + 90.4658845452$$

$$P = 98.0404934806$$

So the price is approximately $\boxed{\mathbf{\$98.04}}$.

4.12

A three-year bond provides a coupon of 8% semiannually and has a cash price of 104. What is the bond's yield?

We know that the coupon pays $\frac{0.08 \cdot 100}{2} = \$4$ due to the semiannual payments. Given that the bond itself therefore pays \$4 every 6 months until maturity, at which it pays \$104, we need to therefore solve

$$4e^{-0.5y} + 4e^{-y} + 4e^{-1.5y} + 4e^{-2y} + 4e^{-2.5y} + 104e^{-3y} = 104$$

The textbook solves this using trial and error, which I would prefer not to, but let's give it a try:

Guess for y	Computed price
0.06	105.163180649
0.065	103.736019541
0.0625	104.447023845
0.0635	104.162005945
0.064	104.019805386
0.06425	103.948782073
0.0641	103.991389907
0.06405	104.005596621

We're extremely close at this point. It seems reasonable enough to say that the rate appears to be very close to 0.06405, or $\boxed{\mathbf{6.405\%}}$.

4.14

Suppose that zero interest rates with continuous compounding are as follows:

Maturity (years)	Rate (% per annum)
1	2.0
2	3.0
3	3.7
4	4.2
5	4.5

Calculate forward interest rates for the second, third, fourth, and fifth years.

We know that the forward rate for the period between T_1 and T_2 implied by today's curve is $R_F = \frac{R_2T_2 - R_1T_1}{T_2 - T_1}$. We are always going year to year in this problem, with each T being a year, so we can further simplify this to just $R_F = R_2T_2 - R_1T_1$.

Year	Forward interest rate
2	$0.03(2) - 0.02(1) = 0.06 - 0.02 = 0.04$, or **4.0%**
3	$0.037(3) - 0.03(2) = 0.111 - 0.06 = 0.051$, or **5.1%**
4	$0.042(4) - 0.037(3) = 0.168 - 0.111 = 0.057$, or **5.7%**
5	$0.045(5) - 0.042(4) = 0.225 - 0.168 = 0.057$, or **5.7%**

4.16

A 10-year, 8% coupon bond currently sells for \$90. A 10-year, 4% coupon bond currently sells for \$80. What is the 10-year zero rate? (Hint: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.)

Using the hint, we long two of the 4% bonds and short one of the 8% bonds. These balance out, with the coupon payment from the longs being $\frac{0.04 \cdot 100}{1} = \$4$ each, so we get $2 \cdot \$4 = \$8$. For the short, we pay $\frac{0.08 \cdot 100}{1} = \$8$. So the values amounts cancel out to 0 dollars each year from coupons. So looking at the actual cash flows, we only consider the initial bond transactions and the final expiry date. For the initial transaction:

$$2 \cdot \text{-\$80} + 1 \cdot \$90 = \text{-\$160 + \$90} = \text{-\$70}$$

And for the expiry date:

$$2 \cdot \$100 - 1 \cdot \$100 = \text{\$200 - \$100} = \$100$$

We treat this as a synthetic zero-coupon as the intermediate cash flows are 0, so we can solve the rate as

$$70 \cdot e^{10(R_c)} = 100$$

$$e^{10(R_c)} = \frac{10}{7}$$

$$10(R_c) = \ln(\frac{10}{7})$$

$$R_c = \frac{\ln(\frac{10}{7})}{10} = \frac{0.35667494393}{10} = 0.035667494393$$

So in conclusion the overall rate is approximately $\boxed{\textbf{3.57\%}}$.

4.22

A five-year bond with a yield of 11% (continuously compounded) pays an 8% coupon at the end of each year.

a)

What is the bond's price?

We know the coupon payment to be $\frac{0.08 \cdot 100}{1} = \$8$. We get the \$100 at the end of the fifth year as well. We therefore can calculate the price as

$$8e^{-0.11(1)} + 8e^{-0.11(2)} + 8e^{-0.11(3)} + 8e^{-0.11(4)} + 108e^{-0.11(5)}$$

$$= 8e^{-0.11} + 8e^{-0.22} + 8e^{-0.33} + 8e^{-0.44} + 108e^{-0.55}$$

$$= 7.16667308237 + 6.4201503837 + 5.75138986746 + 5.15229136867 + 62.310579521$$

$$= 86.8010842233$$

So the bond price here would be approximately $\boxed{\textbf{\$86.80}}$.

b)

What is the bond's duration?

We know the formula for calculating bond duration to be

$$D = \frac{\sum_{i=1}^{n} t_i c_i e^{-yt_i}}{B}$$

$$= \frac{1 \cdot 8 \cdot e^{-0.11(1)} + 2 \cdot 8 \cdot e^{-0.11(2)} + 3 \cdot 8 \cdot e^{-0.11(3)} + 4 \cdot 8 \cdot e^{-0.11(4)} + 5 \cdot 108 \cdot e^{-0.11(5)}}{86.8}$$

$$= \frac{8e^{-0.11} + 16e^{-0.22} + 24e^{-0.33} + 32e^{-0.44} + 540e^{-0.55}}{86.8}$$

$$= \frac{7.16667308237 + 12.8403007674 + 17.2541696024 + 20.6091654747 + 311.552897605}{86.8}$$

$$D = \frac{369.423206532}{86.8} = 4.25602772502$$

So the duration is about $\boxed{\textbf{4.26 years}}$.

c)

Use the duration to calculate the effect on the bond's price of a 0.2% decrease in its yield.

By definition, we know that

$$\Delta B = -BD\Delta y$$

and with $B = 86.8, D = 4.26, \Delta y = -0.002$:

$$\Delta B = -86.8 \cdot 4.26 \cdot -0.002$$

$$\Delta B = 0.739536$$

So the price should $\boxed{\textbf{increase by approximately \$0.74}}$ to \$87.54.

d)

Recalculate the bond's price on the basis of a 10.8% per annum yield and verify that the result is in agreement with your answer to (c).

Using 10.8% instead to calculate the price:

$$8e^{-0.108(1)} + 8e^{-0.108(2)} + 8e^{-0.108(3)} + 8e^{-0.108(4)} + 108e^{-0.108(5)}$$

$$= 8e^{-0.108} + 8e^{-0.216} + 8e^{-0.324} + 8e^{-0.432} + 108e^{-0.540}$$

$$= 7.18102077144 + 6.44588241499 + 5.78600193904 + 5.19367501348 + 62.9368112564$$

$$= 87.5433913953$$

Which is approximately \$87.54, which is indeed $\boxed{\textbf{in agreement with our answer in (c)}}$.

4.34

The following table gives the prices of bonds

Bond Principal (\$)	Time to Maturity (yrs)	Annual Coupon (\$)*	Bond Price (\$)
100	0.5	0.0	98
100	1.0	0.0	95
100	1.5	6.2	101
100	2.0	8.0	104

*Half the stated coupon is paid every six months.

a)

Calculate zero rates for maturities of 6 months, 12 months, 18 months, and 24 months.

We'll use the bootstrap method as described in the text.
For the 6 months, there's no coupon, so the return can be found by solving

$$100 = 98e^{0.5R}$$

$$e^{0.5R} = \frac{100}{98}$$
$$R = 2\ln(1.02040816327)$$
$$R = 0.04040541464$$

so for the 6 month bond, the bond return is approximately $\boxed{\mathbf{4.04\%}}$.

Similarly, for the 12 month, there is no coupon, so we solve accordingly:

$$100 = 95e^{R}$$
$$e^{R} = \frac{100}{95}$$
$$R = \ln(1.05263157895)$$
$$R = 0.05129329439$$

so for the 12 month bond, the return is approximately $\boxed{\mathbf{5.13\%}}$.

For the 18 month, there is a $\frac{\mathbf{\$6.20}}{2} = \$3.10$ coupon every 6 months, with price of \$101, so we modify accordingly and solve:

$$3.10e^{-0.5\cdot 0.0404} + 3.10e^{-1\cdot 0.0513} + 103.10e^{-1.5R} = 101$$
$$3.03800822483 + 2.94498025204 + 103.10e^{-1.5R} = 101$$
$$e^{-1.5R} = \frac{95.0170115231}{103.10}$$
$$-1.5R = \ln(0.92160049973)$$
$$R = 0.05442896452$$

so the 18 month rate is approximately $\boxed{\mathbf{5.44\%}}$.

Finally, the 24 month is similar, with a coupon of $\frac{\$8.00}{2} = \$4.00$ every 6 months, with price of \$104, and we solve similarly:

$$4e^{-0.5\cdot 0.0404} + 4e^{-1\cdot 0.0513} + 4e^{-1.5\cdot 0.0544} + 104e^{-2R} = 104$$
$$4e^{-0.0202} + 4e^{-0.0513} + 4e^{-0.0816} + 104e^{-2R} = 104$$
$$3.9200106127 + 3.79997451876 + 3.68656216477 + 104e^{-2R} = 104$$
$$e^{-2R} = \frac{92.5934527038}{104}$$

$$-2R = \ln(0.89032166061)$$

$$R = 0.05808623256$$

So the 24 month bond has a return of approximately $\boxed{\mathbf{5.81\%}}$

b)

What are the forward rates for the periods: 6 months to 12 months, 12 months to 18 months, 18 months to 24 months?

We know that $R_F = \frac{R_2T_2 - R_1T_1}{T_2 - T_1}$, so we apply accordingly with the rates we've found previously:

- 6 months to 12 months: $\frac{0.0513(1.0)-0.0404(0.5)}{1.0-0.5} = \frac{0.0513-0.0202}{0.5} = 0.0622 = \boxed{\mathbf{6.22\%}}$
- 12 months to 18 months: $\frac{0.0544(1.5)-0.0513(1.0)}{1.5-1.0} = \frac{0.0816-0.0513}{0.5} = 0.0606 = \boxed{\mathbf{6.06\%}}$
- 18 months to 24 months: $\frac{0.0581(2.0)-0.0544(1.5)}{2.0-1.5} = \frac{0.1162-0.0816}{0.5} = 0.0692 = \boxed{\mathbf{6.92\%}}$

c)

What are the 6-month, 12-month, 18-month, and 24-month par yields for bonds that provide semiannual coupon payments?

We know to calculate par yield using $c = \frac{(100-100d)m}{A}$, with $m = 2, d = e^{-R_nT_n}$, and A is the cumulative sums of the previous d . So for each bond, we calculate cumulative A and then solve for c.

- 6 month:

$$c = \frac{2(100 - 100 \cdot e^{-0.0404 \cdot 0.5})}{e^{-0.040 \cdot 0.5}}$$

$$= \frac{2(100 - 100 \cdot e^{-0.0202})}{e^{-0.0202}}$$

$$= \frac{2(100 - 100 \cdot 0.98000265317)}{0.98000265317}$$

$$= \frac{3.999469366}{0.98000265317}$$

$$c = 4.08108014102$$

so the 6 month par yield is approximately $\boxed{\mathbf{4.08\%}}$

- 12 month:

$$c = \frac{2(100 - 100 \cdot e^{-0.0513 \cdot 1.0})}{e^{-0.040 \cdot 0.5} + e^{-0.0513 \cdot 1.0}}$$
$$= \frac{2(100 - 100 \cdot e^{-0.0513})}{e^{-0.0202} + e^{-0.0513}}$$
$$= \frac{2(100 - 100 \cdot 0.94999362969)}{0.98000265317 + 0.94999362969}$$
$$= \frac{10.001274062}{1.92999628286}$$
$$c = 5.18201726647$$

so the 12 month par yield is approximately $\boxed{\mathbf{5.18\%}}$

- 18 month:

$$c = \frac{2(100 - 100 \cdot e^{-0.0544 \cdot 1.5})}{e^{-0.040 \cdot 0.5} + e^{-0.0513 \cdot 1.0} + e^{-0.0544 \cdot 1.5}}$$
$$= \frac{2(100 - 100 \cdot e^{-0.0816})}{e^{-0.0202} + e^{-0.0513} + e^{-0.0816}}$$
$$= \frac{2(100 - 100 \cdot 0.92164054119)}{0.98000265317 + 0.94999362969 + 0.92164054119}$$
$$= \frac{15.671891762}{2.85163682405}$$
$$c = 5.49575304605$$

so the 18 month par yield is approximately $\boxed{\mathbf{5.50\%}}$

- 24 month:

$$c = \frac{2(100 - 100 \cdot e^{-0.0581 \cdot 2.0})}{e^{-0.040 \cdot 0.5} + e^{-0.0513 \cdot 1.0} + e^{-0.0544 \cdot 1.5} + e^{-0.0581 \cdot 2.0}}$$
$$= \frac{2(100 - 100 \cdot e^{-0.1162})}{e^{-0.0202} + e^{-0.0513} + e^{-0.0816} + e^{-0.1162}}$$
$$= \frac{2(100 - 100 \cdot 0.89029714606)}{0.98000265317 + 0.94999362969 + 0.92164054119 + 0.89029714606}$$
$$= \frac{21.940570788}{3.74193397011}$$
$$c = 5.86343077223$$

so the 24 month par yield is approximately $\boxed{\mathbf{5.86\%}}$

d)

Estimate the price and yield of a two-year bond providing a semiannual coupon of 7% per annum.

We know each coupon payment to be $\frac{0.07 \cdot 100}{2}$ =\$3.50. We calculate accordingly:

$$P = 3.5e^{-0.0404(0.5)} + 3.5e^{-0.0513(1.0)} + 3.5e^{-0.0544(1.5)} + 103.5e^{-0.0581(2.0)}$$

$$= 3.5e^{-0.0202} + 3.5e^{-0.0513} + 3.5e^{-0.0816} + 103.5e^{-0.1162}$$

$$= 3.4300092861 + 3.32497770392 + 3.22574189417 + 92.1457546172$$

$$P = 102.126483501$$

so the price is $\boxed{\mathbf{\$102.13}}$. For the yield, let's solve using this new price:

$$3.5e^{-0.5y} + 3.5e^{-y} + 3.5e^{-1.5y} + 103.5e^{-2y} = 102.13$$

$$3.5e^{-0.5y} + 3.5e^{-y} + 3.5e^{-1.5y} + 103.5e^{-2y} = 102.13$$

We can guess and check for values of y as we did before, but instead I'll use an external solver to find $y \approx 0.0577183$, so we find the yield to be approximately $\boxed{\mathbf{5.77\%}}$.